SET-M

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B.SC.[ECS] - II Sem III Practical Examination, October 2026

VSC1: Based on DSC1-3 and DSC1-4

Date & Time: _____ Max Marks: 30

Instructions:

I) Attempt any one question from each section.

II) Each question carries 10 marks.

III) The Journal and Viva-Voce carry 5 marks.

Section-I: Based on DSC1-3 (Data Structures using C++)

Q.1 Write a program to implement insert, remove and display operations of queue using linked list.

Q.2 Write a program to implement sequential search method for unsorted data.

Section-II: Based on DSC1-4 (Database Management Systems)

Q.3 Create a table book by applying proper constraints, insert at least 5 records and solve the following queries:

Field Name	Data Type	Constraint
Book_ID	INT	Primary Key
Book_Name	VARCHAR(100)	NOT NULL
Author	VARCHAR(50)	-
Publisher	VARCHAR(50)	-
Category	VARCHAR(30)	-
Price	DECIMAL(8,2)	-
Quantity	INT	-

a. Find the highest-priced and lowest-priced books.

b. Calculate the total value of all books.

c. Count the total number of books recorded in the table.

d. Increase the price of all books by 10%.

e. Delete a book using its Book_ID.

Q.4 Create a table by applying proper constraints, insert at least 5 records and solve the following queries:

A] Customer:

Field Name	Data Type	Constraint
Customer_ID	INT	Primary Key
Customer_Name	VARCHAR(50)	NOT NULL
City	VARCHAR(30)	-
Phone	VARCHAR(15)	-

B] Order:

Field Name	Data Type	Constraint
Order_ID	INT	Primary Key
Customer_ID	INT	Foreign Key referencing Customer(Customer_ID)
Order_Date	DATE	-
Amount	DECIMAL(10,2)	-
Status	VARCHAR(20)	-

a. Display the total order amount for each customer.

b. Display customers who have placed more orders than the average number of orders per customer.

c. Display customers who have placed orders with status 'Pending'.

d. Display the customer or customers with the maximum total order amount.

e. Update the status of a specified order.